Full results for small datasets

Overview datasets

Dataset	n	d	Type	Source
Pulsar-2k	2,000	8	Tabular	UCI ML Repository
Abalone-2k	2,000	10	Tabular	UCI ML Repository
Electric-2k	2,000	12	Tabular	UCI ML Repository
Facebook-2k	2,000	13	Tabular	UCI ML Repository
Frogs-2k	2,000	22	Tabular	UCI ML Repository
Creditcard-2k	2,000	24	Tabular	UCI ML Repository
Travel-2k	2,000	24	Tabular	UCI ML Repository
Bank-2k	2,000	53	Tabular	UCI ML Repository
Plants-2k	2,000	70	Tabular	UCI ML Repository
Adult-2k	2,000	110	Tabular	UCI ML Repository

Metric: Cityblock (L1, Manhattan)

Objective function values

					Deviation from optimal objective (%)								
Dataset	n	d	Optimal objective	UB gap (%)	GRB	HEX-B	NWX	BLV	ABM	ABM-i5	AC	GRD	RND
Pulsar-2k	2,000	8	8,270.66	-0.05	0.00	0.00	0.00	0.00	0.02	0.00	0.03	1.48	24.25
Abalone-2k	2,000	10	15,483.73	-0.02	0.00	0.00	0.00	0.00	0.02	0.00	0.06	0.15	30.52
Electric-2k	2,000	12	20,433.84	-0.35	0.00	0.00	0.00	0.00	0.76	0.10	1.75	1.63	32.24
Facebook-2k	2,000	13	8,045.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7.37	22.81
Frogs-2k	2,000	22	35,212.07	-3.04	0.00	0.00	0.00	0.00	0.14	0.01	0.36	3.82	27.50
Creditcard-2k	2,000	24	27,092.41	-0.10	0.00	0.00	0.00	0.00	0.08	0.01	0.29	3.37	26.16
Travel-2k	2,000	24	32,827.94	-1.09	0.00	0.00	0.00	0.00	0.18	0.01	0.74	3.45	25.24
Bank-2k	2,000	53	21,231.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	1.52	21.97
Plants-2k	2,000	70	19,284.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.82	16.13
Adult-2k	2,000	110	46,209.57	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	1.39	17.48
Average					0.00	0.00	0.00	0.00	0.12	0.01	0.33	2.90	24.43

Runtimes (seconds)

			Runtime (s)								
Dataset	n	d	GRB	HEX-B	NWX	BLV	ABM	ABM-i5	AC	GRD	RND
Pulsar-2k	2,000	8	5.41	464.35	2,273.12	69.23	0.22	1.30	2.69	1.79	<0.01
Abalone-2k	2,000	10	4.76	429.04	2,135.25	51.39	0.20	1.28	2.69	1.73	<0.01
Electric-2k	2,000	12	5.37	423.66	1,683.40	7.08	0.07	0.40	2.69	1.78	<0.01
Facebook-2k	2,000	13	6.53	429.16	2,424.31	63.41	0.24	1.48	2.69	1.69	<0.01
Frogs-2k	2,000	22	5.61	419.83	2,035.88	51.02	0.17	1.02	2.67	1.67	<0.01
Creditcard-2k	2,000	24	6.70	441.22	2,117.90	44.55	0.18	1.24	2.69	1.79	<0.01
Travel-2k	2,000	24	6.03	437.07	1,766.03	22.93	0.14	0.84	2.68	1.71	<0.01
Bank-2k	2,000	53	6.00	442.89	2,059.15	68.74	0.19	1.20	2.68	1.79	<0.01
Plants-2k	2,000	70	4.00	335.71	518.84	2.72	0.04	0.19	2.69	1.43	<0.01
Adult-2k	2,000	110	5.54	450.64	2,916.55	56.93	0.32	2.10	2.70	1.81	<0.01
Average			5.60	427.36	1,993.04	43.80	0.18	1.10	2.69	1.72	<0.01

Metric: Euclidean (L2)

Objective function values

					Deviation from optimal objective (%)								
Dataset	n	d	Optimal objective	UB gap (%)	GRB	HEX-B	NWX	BLV	ABM	ABM-i5	AC	GRD	RND
Pulsar-2k	2,000	8	3,773.79	-0.67	0.00	0.00	0.00	0.00	0.11	0.01	0.20	6.76	24.24
Abalone-2k	2,000	10	5,556.83	-0.92	0.00	0.00	0.00	0.00	0.04	0.01	0.13	1.12	27.10
Electric-2k	2,000	12	6,568.55	-0.17	0.00	0.00	0.00	0.00	0.83	0.11	1.69	1.59	26.61
Facebook-2k	2,000	13	4,690.48	-1.13	0.00	0.00	0.00	0.00	0.00	0.00	0.05	18.43	22.96
Frogs-2k	2,000	22	8,943.46	-3.94	0.00	0.00	0.00	0.00	0.14	0.01	0.39	3.27	25.41
Creditcard-2k	2,000	24	7,792.36	-1.14	0.00	0.00	0.00	0.00	0.23	0.03	0.58	7.71	21.12
Travel-2k	2,000	24	8,580.14	-1.03	0.00	0.00	0.00	0.00	0.21	0.02	0.66	3.72	18.86
Bank-2k	2,000	53	7,714.37	-2.56	0.00	0.00	0.00	0.00	0.05	0.01	0.13	6.16	13.14
Plants-2k	2,000	70	4,086.04	-1.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.98	12.93
Adult-2k	2,000	110	14,118.25	-2.01	0.00	0.00	0.00	0.00	0.06	0.01	0.16	6.90	9.01
Average					0.00	0.00	0.00	0.00	0.17	0.02	0.40	6.76	20.14

Runtimes (seconds)

			Runtime (s)								
Dataset	n	d	GRB	HEX-B	NWX	BLV	ABM	ABM-i5	AC	GRD	RND
Pulsar-2k	2,000	8	5.67	442.98	1,972.89	29.87	0.23	1.28	2.69	1.77	<0.01
Abalone-2k	2,000	10	6.18	446.18	1,898.18	78.59	0.17	1.00	2.69	1.71	<0.01
Electric-2k	2,000	12	7.45	505.92	1,602.58	4.36	0.08	0.43	2.69	1.79	<0.01
Facebook-2k	2,000	13	8.52	450.94	2,606.83	73.68	0.24	1.60	2.68	1.64	<0.01
Frogs-2k	2,000	22	5.81	415.11	1,859.11	56.82	0.17	1.00	2.67	1.66	<0.01
Creditcard-2k	2,000	24	5.74	435.77	1,836.54	14.18	0.18	1.00	2.70	1.78	<0.01
Travel-2k	2,000	24	6.05	440.15	1,758.04	22.85	0.15	0.84	2.68	1.71	<0.01
Bank-2k	2,000	53	6.22	443.64	2,022.93	23.07	0.21	1.30	2.69	1.78	<0.01
Plants-2k	2,000	70	4.09	379.46	2,448.64	33.86	0.28	1.61	2.69	1.41	<0.01
Adult-2k	2,000	110	5.59	446.59	2,066.80	15.39	0.29	1.49	2.69	1.80	<0.01
Average			6.13	440.67	2,007.26	35.27	0.20	1.16	2.69	1.71	<0.01

Metric: Squared Euclidean (L2 squared)

Objective function values

					Deviation from optimal objective (%)									
Dataset	n	d	Optimal objective	UB gap (%)	GRB	HEX-B	HEX-L	NWX	BLV	ABM	ABM-i5	AC	GRD	RND
Pulsar-2k	2,000	8	20,119.86	-4.24	0.00	0.00	0.06	0.00	0.00	0.24	0.03	0.36	3.89	36.82
Abalone-2k	2,000	10	36,229.74	-0.82	0.00	0.00	0.02	0.00	0.00	0.07	0.01	0.20	0.73	43.70
Electric-2k	2,000	12	43,820.25	-0.35	0.00	0.00	1.17	0.00	0.00	1.70	0.21	3.12	2.28	45.06
Facebook-2k	2,000	13	28,936.39	-7.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	5.81	19.52
Frogs-2k	2,000	22	81,833.59	-4.89	0.00	0.00	0.13	0.00	0.00	0.30	0.03	0.71	3.84	37.26
Creditcard-2k	2,000	24	69,188.42	-1.92	0.00	0.00	0.22	0.00	0.00	0.35	0.04	0.78	6.97	32.15
Travel-2k	2,000	24	75,166.04	-1.50	0.00	0.00	0.34	0.00	0.00	0.39	0.04	1.20	4.64	32.26
Bank-2k	2,000	53	61,541.42	-3.60	0.00	0.00	0.02	0.00	0.00	0.07	0.01	0.11	2.49	18.83
Plants-2k	2,000	70	19,284.00	-0.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.82	16.13
Adult-2k	2,000	110	235,968.78	-0.30	0.00	0.00	0.05	0.00	0.00	0.05	0.01	0.12	1.85	9.84
Average					0.00	0.00	0.20	0.00	0.00	0.32	0.04	0.66	3.73	29.16

Runtimes (seconds)

			Runtime (s)									
Dataset	n	d	GRB	HEX-B	HEX-L	NWX	BLV	ABM	ABM-i5	AC	GRD	RND
Pulsar-2k	2,000	8	5.96	450.17	7,200.23	1,729.14	26.17	0.16	0.90	6.77	1.77	<0.01
Abalone-2k	2,000	10	6.23	452.84	7,199.89	1,727.65	69.87	0.16	0.93	7.60	1.76	<0.01
Electric-2k	2,000	12	4.88	423.64	7,200.13	1,631.54	5.41	0.07	0.38	9.20	1.77	<0.01
Facebook-2k	2,000	13	6.55	454.14	7,200.02	2,260.27	97.77	0.22	1.51	10.38	1.67	<0.01
Frogs-2k	2,000	22	5.86	415.50	7,199.88	1,867.74	50.30	0.16	0.92	16.04	1.65	<0.01
Creditcard-2k	2,000	24	6.20	437.55	7,200.00	1,793.01	13.86	0.16	0.95	16.65	1.77	<0.01
Travel-2k	2,000	24	5.70	439.71	7,199.81	1,700.48	20.87	0.14	0.77	16.33	1.70	<0.01
Bank-2k	2,000	53	6.62	440.74	7,200.10	2,062.01	34.46	0.20	1.30	34.33	1.78	<0.01
Plants-2k	2,000	70	3.96	336.08	7,200.92	529.61	2.69	0.04	0.20	42.30	1.42	<0.01
Adult-2k	2,000	110	6.01	441.54	7,201.75	2,124.53	17.77	0.28	1.56	67.15	1.82	<0.01
Average			5.80	429.19	7,200.27	1,742.60	33.92	0.16	0.94	22.67	1.71	<0.01